Mentify

AI Enhanced Digital Learning Platform

Software Requirement Specification (SRS)

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01	2026/03/14	Project Team	Initial draft created

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Chapter 1: Introduction

The Mentify Digital Learning Platform is a comprehensive Learning Management System (LMS) designed and developed for TIT Education, an educational institution that provides tailored tuition services for students. This document presents the complete Software Requirement Specification (SRS) for the proposed system, capturing all relevant functional and non-functional requirements, use cases, activity flows, system constraints, and feasibility considerations.

Furthermore, this document has been prepared in accordance with the guidelines of the SENG 31242 module and adheres to the IEEE standard for software requirement documentation.

1.1 Purpose

The purpose of this document is to provide a detailed and structured specification of the requirements for the Mentify Digital Learning Platform. It is intended to serve as the primary reference for the development team, the client (TIT Education), project supervisors, and all stakeholders involved in the design, development, testing, and deployment of the system. This document defines the system's functional capabilities, non-functional quality attributes, user interactions, business rules, and technical constraints.

The intended audience of this document includes software engineers, UI/UX designers, quality assurance engineers, the project supervisor, and the client representative at TIT Education.

1.2 Scope

System Name: Mentify – AI Enhanced Digital Learning Platform

The Mentify platform is a web-based Learning Management System (LMS) designed to deliver AI-powered personalised learning experiences for students. The system enables students to enrol in courses, access multimedia learning content, complete assessments, communicate with teachers and peers, and track their academic progress through visual dashboards.

Teachers are provided with functionalities to create and manage courses, upload learning materials, design quizzes and assignments, evaluate student submissions with AI-assisted grading, and monitor student performance through analytical tools. Administrators oversee the overall operation of the platform, including user management, role-based access control, course oversight, and system health monitoring. A dedicated Support Team manages user support tickets, maintains a knowledge base, and escalates technical issues when required.

An integrated AI subsystem supports content summarisation, quiz generation, automated grading, personalised course recommendations, and provides a 24/7 AI-powered help chatbot to enhance the learning experience.

Explicit Exclusions:

The system shall not include live video conferencing or real-time virtual classroom functionality in the initial release. Payment processing and financial management modules are excluded from the current scope. Offline access to course content will not be supported in the first version of the system. Integration with external Student Information Systems (SIS) of other institutions is not considered within the scope of this implementation.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
LMS	Learning Management System, a software platform used for delivering, managing, and tracking educational content.
AI	Artificial Intelligence, a technology that enables automated decision-making, content generation, and personalisation.
JWT	JSON Web Token, a compact and URL-safe token used for secure authentication between client and server.
SSO	Single Sign-On, an authentication mechanism that allows users to log in once and access multiple systems.
MFA	Multi-Factor Authentication, a security method that requires two or more verification factors during login.
RBAC	Role-Based Access Control, a method of restricting system access based on user roles.
MCQ	Multiple Choice Question, a question format consisting of predefined answer options.
SRS	Software Requirement Specification, a document that describes the functional and non-functional requirements of a system.
NFR	Non-Functional Requirement, a requirement that specifies system quality attributes such as performance, security, and usability.
FAQ	Frequently Asked Questions, a collection of commonly asked questions and their corresponding answers.
PDF	Portable Document Format, a file format used to present documents independently of software or hardware.
API	Application Programming Interface, a set of protocols used for building and integrating application software.
UI	User Interface, the visual elements through which a user interacts with the system.
CRUD	Create, Read, Update, Delete, the four fundamental operations of persistent data storage.
HTTPS	Hypertext Transfer Protocol Secure, an encrypted version of HTTP used for secure data transmission.

TLS	Transport Layer Security, a cryptographic protocol used to secure communications over a network.
SMTP	Simple Mail Transfer Protocol, a standard protocol used for sending emails.

Table 2: Definitions, Acronyms and Abbreviations

1.4 Document Overview

This document is organised into multiple chapters that collectively define the Software Requirement Specification (SRS) for the Mentify Digital Learning Platform. Chapter 1 provides the introductory information, including the purpose, scope, definitions, and overall document structure. Chapter 2 describes the project scope in detail and identifies the key stakeholders involved in the system.

Chapter 3 presents the functional requirements of the system, while Chapter 4 specifies the non-functional requirements aligned with recognised quality attributes. Chapter 5 outlines the applicable business rules, system constraints, and underlying assumptions. Chapters 6 and 7 include the use case diagrams, detailed use case descriptions, and activity diagrams that illustrate system interactions and workflows.

Chapter 8 describes the external interface requirements, including user, hardware, software, and communication interfaces. Chapter 9 defines the system's data requirements. Chapter 10 discusses alternative solutions considered during the system design phase. Chapter 11 presents the feasibility study covering technical, economic, operational, and schedule feasibility aspects. Chapter 12 provides a detailed risk analysis. Finally, Chapter 13 lists all references used in the preparation of this document.

Chapter 2: Project Scope and Stakeholders

2.1 Project Scope

The Mentify Digital Learning Platform aims to transform TIT Education's existing manual and static educational processes into a modern, AI-driven, student-centred digital learning environment. The platform shall serve five categories of users: Students, Teachers, Administrators, Support Team members, and an integrated AI System.

The system shall deliver AI-powered personalised learning experiences by analysing individual student behaviour, progress, and preferences to dynamically adapt content, recommendations, and learning paths tailored to each learner's needs. It shall automate assessment, feedback, and content management processes by replacing manual workflows with intelligent automated grading, instant feedback generation, and streamlined course management tools for educators.

Furthermore, the platform shall provide a centralised collaboration and communication environment that connects students and educators through real-time messaging, discussion forums, and analytics-driven performance insights, thereby extending the learning experience beyond the traditional classroom setting. Although the system is initially developed for TIT Education, it is designed with scalability considerations to support future expansion to the broader student community.

2.2 Stakeholders

Stakeholder	Role	Interest / Responsibility
TIT Education (Client)	Project sponsor & primary beneficiary	Approves requirements, provides domain knowledge, and funds the project
Students	Primary end users	Use the platform for learning, assessments, communication, and progress tracking
Teachers	Primary end users	Create courses, deliver content, assess students, and monitor performance

System Administrators	Internal operators	Manage users, roles, courses, system configuration, and overall platform health
Support Team	Internal operators	Resolve user issues, manage support tickets, and maintain the knowledge base
Development Team (Mentify)	System builders	Design, develop, test, and deploy the platform
Project Supervisor	Academic oversight	Provides guidance to the development team and evaluates project deliverables

Table 3: Stakeholder Register

Chapter 3: Functional Requirements

This chapter specifies the functional requirements of the Mentify Digital Learning Platform. These requirements describe the expected system behaviours and operations for Students, Teachers, Administrators, Support Team members, and the integrated AI subsystem.

3.1 Student Functional Requirements

ID	Description	Priority	Source	Status
FR-S001	The system shall allow a new student to register by entering their name, email, grade, and password, or by using Google authentication.	High	Student, Admin	Proposed
FR-S002	The system shall allow students to log in using JWT-based secure authentication.	High	Student	Proposed
FR-S003	The system shall allow students to view and edit their profile including photo and email.	Medium	Student	Proposed
FR-S004	The system shall allow students to change their account password.	Medium	Student	Proposed
FR-S005	The system shall allow students to log out with secure session termination.	High	Student	Proposed
FR-S006	The system shall allow students or parents to view and update parental contact information.	Medium	Student, Parent	Proposed

FR-S007	The system shall allow students to view all available courses with filtering by grade level.	High	Student	Proposed
FR-S008	The system shall allow students to enrol in a course.	High	Student	Proposed
FR-S009	The system shall allow students to view a list of all their enrolled courses.	High	Student	Proposed
FR-S010	The system shall allow students to drop (unenrol from) a course.	Medium	Student	Proposed
FR-S011	The system shall provide AI-based course recommendations by analysing the student's learning history.	Medium	Student, AI System	Proposed
FR-S012	The system shall display a course dashboard with an overview of modules, lessons, progress, and announcements.	High	Student	Proposed
FR-S013	The system shall allow students to download study materials uploaded by teachers.	High	Student	Proposed
FR-S014	The system shall allow students to take and save personal notes within the platform.	Low	Student	Proposed
FR-S015	The system shall allow students to	Medium	Student	Proposed

	search topics by keyword or chapter across their courses.			
FR-S016	The system shall allow students to mark individual lessons as completed.	Medium	Student	Proposed
FR-S017	The system shall provide an AI learning assistant that can summarise content, answer questions, and explain difficult topics.	Medium	Student, AI System	Proposed
FR-S018	The system shall allow students to view lesson videos and multimedia content.	High	Student	Proposed
FR-S019	The system shall allow students to bookmark lessons for quick access.	Low	Student	Proposed
FR-S020	The system shall display all assignments and their due dates to students.	High	Student	Proposed
FR-S021	The system shall allow students to upload and submit completed assignments.	High	Student	Proposed
FR-S022	The system shall allow students to attempt online quizzes including MCQ and short answer formats.	High	Student	Proposed
FR-S023	The system shall	High	Student	Proposed

	display quiz results including marks and correct solutions after completion.			
FR-S024	The system shall allow students to view all their grades and scores across assignments and quizzes.	High	Student	Proposed
FR-S025	The system shall allow students to view feedback provided by the teacher or generated by the AI.	Medium	Student, Teacher, AI System	Proposed
FR-S026	The system shall allow students to chat with classmates in group chats within a course.	Medium	Student	Proposed
FR-S027	The system shall allow students to send private messages to teachers.	Medium	Student, Teacher	Proposed
FR-S028	The system shall allow students to participate in topic-based discussion forums within a course.	Medium	Student	Proposed
FR-S029	The system shall provide a 24/7 AI-powered help chatbot for study questions, platform issues, and course content.	Medium	Student, AI System	Proposed
FR-S030	The system shall deliver notifications to students about	High	Student	Proposed

	new assignments, announcements, grades, and messages.			
FR-S031	The system shall display visual progress reports with charts showing completion, scores, and learning trends.	Medium	Student	Proposed
FR-S032	The system shall allow students to set personal learning goals.	Low	Student	Proposed
FR-S033	The system shall allow students to view their attendance records.	Medium	Student	Proposed

3.2 Teacher Functional Requirements

ID	Description	Priority	Source	Status
FR-T001	The system shall allow a new teacher to register by entering personal details, subject specialisation, and credentials.	High	Teacher, Admin	Proposed
FR-T002	The system shall allow teachers to log in using JWT-based secure authentication.	High	Teacher	Proposed
FR-T003	The system shall allow teachers to	Medium	Teacher	Proposed

	view and edit their profile information.			
FR-T004	The system shall allow teachers to reset their password via a "forgot password" mechanism.	Medium	Teacher	Proposed
FR-T005	The system shall allow teachers to log out with secure session termination.	High	Teacher	Proposed
FR-T006	The system shall allow teachers to create a new course with name, description, and grade level.	High	Teacher	Proposed
FR-T007	The system shall allow teachers to edit or delete an existing course.	High	Teacher	Proposed
FR-T008	The system shall allow teachers to view the list of students enrolled in each of their courses.	High	Teacher	Proposed
FR-T009	The system shall allow teachers to approve or reject student enrolment requests.	Medium	Teacher	Proposed

FR-T010	The system shall allow teachers to update course contents and materials.	High	Teacher	Proposed
FR-T011	The system shall allow teachers to set scheduled time slots for class sessions.	Medium	Teacher	Proposed
FR-T012	The system shall allow teachers to manage and update their availability schedule.	Medium	Teacher	Proposed
FR-T013	The system shall allow teachers to upload learning materials including PDFs, videos, and slides.	High	Teacher	Proposed
FR-T014	The system shall allow teachers to organise lessons into modules, chapters, or units.	High	Teacher	Proposed
FR-T015	The system shall allow teachers to edit or delete individual lessons.	Medium	Teacher	Proposed
FR-T016	The system shall allow teachers to schedule lesson	Medium	Teacher	Proposed

	release dates.			
FR-T017	The system shall allow teachers to view student engagement data including lesson views and study time.	Medium	Teacher	Proposed
FR-T018	The system shall allow teachers to add subtitles or transcripts to video lessons.	Low	Teacher	Proposed
FR-T019	The system shall allow teachers to set prerequisite lessons.	Medium	Teacher	Proposed
FR-T020	The system shall allow teachers to post course announcements to enrolled students.	Medium	Teacher	Proposed
FR-T021	The system shall allow teachers to create assignments with instructions, file requirements, and deadlines.	High	Teacher	Proposed
FR-T022	The system shall allow teachers to view all submitted assignments.	High	Teacher	Proposed
FR-T023	The system shall	High	Teacher	Proposed

	allow teachers to grade student assignments.			
FR-T024	The system shall allow teachers to create online quizzes with questions, answer choices, and scoring rules.	High	Teacher	Proposed
FR-T025	The system shall provide AI auto-generation of quiz questions based on course content.	Medium	Teacher, AI System	Proposed
FR-T026	The system shall allow teachers to publish quiz results to students.	High	Teacher	Proposed
FR-T027	The system shall provide AI-assisted grading for assignments and quizzes.	Medium	Teacher, AI System	Proposed
FR-T028	The system shall allow teachers to set assignment deadlines and submission rules.	High	Teacher	Proposed
FR-T029	The system shall allow teachers to return graded assignments with	High	Teacher	Proposed

	written feedback.			
FR-T030	The system shall allow teachers to chat with individual students or groups.	Medium	Teacher	Proposed
FR-T031	The system shall allow teachers to post class-wide announcements.	Medium	Teacher	Proposed
FR-T032	The system shall allow teachers to moderate discussion forums.	Medium	Teacher	Proposed
FR-T033	The system shall allow teachers to send bulk notifications to all students in a course.	Medium	Teacher	Proposed
FR-T034	The system shall display a student performance dashboard with scores, completion rates, and progress.	High	Teacher	Proposed
FR-T035	The system shall allow teachers to manually trigger notifications and alerts to students.	Medium	Teacher	Proposed
FR-T036	The system shall provide AI-generated insights	Medium	Teacher, AI System	Proposed

	identifying student weaknesses.			
FR-T037	The system shall allow teachers to export student performance reports.	Medium	Teacher	Proposed
FR-T038	The system shall allow teachers to view attendance reports.	Medium	Teacher	Proposed

3.3 Administrator Functional Requirements

ID	Description	Priority	Source	Status
FR-A001	The system shall allow the admin to create and edit teacher and student accounts.	High	Admin	Proposed
FR-A002	The system shall allow the admin to assign or modify user roles (Admin, Teacher, Student).	High	Admin	Proposed
FR-A003	The system shall allow the admin to deactivate or archive user accounts.	Medium	Admin	Proposed
FR-A004	The system shall allow the admin to configure Single Sign-On (SSO).	Medium	Admin	Proposed
FR-A005	The system shall allow the admin to	Medium	Admin	Proposed

	configure Multi-Factor Authentication (MFA).			
FR-A006	The system shall allow the admin to configure Role-Based Access Control (RBAC).	High	Admin	Proposed
FR-A007	The system shall allow the admin to create custom roles beyond the default three.	Low	Admin	Proposed
FR-A008	The system shall allow the admin to view all courses and content on the platform.	High	Admin	Proposed
FR-A009	The system shall allow the admin to audit course content for compliance with institutional standards.	Medium	Admin	Proposed
FR-A010	The system shall allow the admin to publish or unpublish courses.	High	Admin	Proposed
FR-A011	The system shall allow the admin to manage course	Medium	Admin	Proposed

	categories and tags.			
FR-A012	The system shall allow the admin to assign a teacher to a course.	High	Admin	Proposed
FR-A013	The system shall allow the admin to generate comprehensive usage reports.	Medium	Admin	Proposed
FR-A014	The system shall allow the admin to analyse system engagement metrics.	Medium	Admin	Proposed

3.4 Support Team Functional Requirements

ID	Description	Priority	Source	Status
FR-SP001	The system shall allow the support team to view and filter support tickets by status, priority, or type.	High	Support Team	Proposed
FR-SP002	The system shall allow the support team to assist users with login and password issues.	High	Support Team	Proposed
FR-SP003	The system shall allow the support team to guide users through platform	Medium	Support Team	Proposed

	features.			
FR-SP004	The system shall allow the support team to troubleshoot technical glitches.	High	Support Team	Proposed
FR-SP005	The system shall allow the support team to provide course enrolment support.	Medium	Support Team	Proposed
FR-SP006	The system shall allow the support team to create and edit FAQ articles.	Medium	Support Team	Proposed
FR-SP007	The system shall allow the support team to organise knowledge base categories.	Medium	Support Team	Proposed
FR-SP008	The system shall allow the support team to identify gaps in knowledge base content.	Low	Support Team	Proposed
FR-SP009	The system shall allow the support team to escalate complex issues to admins or developers.	High	Support Team	Proposed
FR-SP010	The system shall allow the support	Medium	Support Team	Proposed

	team to collaborate with teachers on student issues.			
FR-SP011	The system shall provide the support team with read-only access to relevant admin tools.	Medium	Support Team	Proposed

3.5 AI System Functional Requirements

ID	Description	Priority	Source	Status
FR-AI001	The AI system shall analyse student learning history and provide personalised course recommendations.	Medium	AI System, Student	Proposed
FR-AI002	The AI system shall summarise lesson content and explain difficult topics on demand.	Medium	AI System, Student	Proposed
FR-AI003	The AI system shall provide a 24/7 chatbot for answering study questions, platform issues, and course content queries.	Medium	AI System, Student	Proposed
FR-AI004	The AI system shall auto-generate quiz questions based on course content.	Medium	AI System, Teacher	Proposed

FR-AI005	The AI system shall assist in grading assignments and quizzes, particularly objective-type questions.	Medium	AI System, Teacher	Proposed
FR-AI006	The AI system shall analyse student performance data and identify specific areas of weakness.	Medium	AI System, Teacher	Proposed
FR-AI007	The AI system shall generate feedback on student submissions.	Medium	AI System, Teacher, Student	Proposed

Table 4: Functional Requirements

Chapter 4: Non-Functional Requirements

Non-functional requirements define the quality attributes, performance expectations, operational constraints, and system characteristics that the Mentify Digital Learning Platform must satisfy.

4.1 Performance

ID	Category	Description	Acceptance Criterion
NFR-P001	Performance	The system shall respond to user interactions promptly under normal load conditions.	95% of page load requests shall complete within 2 seconds under normal load (up to 500 concurrent users).
NFR-P002	Performance	The system shall handle search queries efficiently.	95% of search queries shall return results within 2 seconds.
NFR-P003	Performance	The system shall process quiz submissions without delay.	Quiz submission and result calculation shall complete within 3 seconds.
NFR-P004	Performance	The system shall support concurrent usage during peak hours.	The system shall support a minimum of 1,000 concurrent users without response time degradation exceeding 20%.

4.2 Reliability

ID	Category	Description	Acceptance Criterion
NFR-R001	Reliability	The system shall maintain high availability.	The system shall achieve 99.5% uptime measured monthly, excluding scheduled maintenance.
NFR-R002	Reliability	The system shall recover from failures with minimal data loss.	Full recovery shall occur within 30 minutes with no more than 5 minutes of transactional data loss.
NFR-R003	Reliability	The system shall perform automated backups.	Automated database backups shall occur every 24 hours and be

			stored in a geographically separate location.

4.3 Usability

ID	Category	Description	Acceptance Criterion
NFR-U001	Usability	The system shall be easy to learn for new users.	A new student shall enrol in a course and access lesson content within 5 minutes of first login without assistance.
NFR-U002	Usability	The system shall provide a responsive user interface.	The UI shall function correctly on screens ranging from 320px to 1920px.
NFR-U003	Usability	The system shall meet accessibility standards.	The system shall comply with WCAG 2.1 Level AA guidelines.
NFR-U004	Usability	The system shall support multiple languages.	The system shall support English and Tamil as primary interface languages.

4.4 Security

ID	Category	Description	Acceptance Criterion
NFR-SE001	Security	The system shall authenticate users securely.	Authentication shall use JWT tokens with a maximum expiry of 24 hours and refresh token support.
NFR-SE002	Security	The system shall encrypt sensitive data.	Data in transit shall use TLS 1.2 or higher; passwords shall be hashed using bcrypt (minimum cost factor 10).
NFR-SE003	Security	The system shall enforce role-based access control.	Users shall not access functionality beyond assigned permissions, verified via RBAC middleware.
NFR-SE004	Security	The system shall protect against	The system shall pass OWASP Top 10

		common web vulnerabilities.	assessment with no critical or high-severity findings.
NFR-SE005	Security	The system shall support Multi-Factor Authentication.	MFA shall be available for all roles and mandatory for admin accounts.

4.5 Scalability

ID	Category	Description	Acceptance Criterion
NFR-SC001	Scalability	The system shall scale horizontally to accommodate growth.	The architecture shall support a 300% user base increase without redesign.
NFR-SC002	Scalability	The system shall handle growing data volumes.	The database shall manage up to 1 million course records and 10 million activity records with performance degradation not exceeding 10%.

4.6 Maintainability

ID	Category	Description	Acceptance Criterion
NFR-M001	Maintainability	The system shall follow modular architecture principles.	The system shall adopt a microservice architecture enabling independent module deployment.
NFR-M002	Maintainability	The system shall maintain comprehensive code documentation.	All APIs shall include Swagger/OpenAPI documentation; code coverage shall exceed 80%.
NFR-M003	Maintainability	The system shall support continuous integration and deployment.	Automated CI/CD pipelines shall enable deployment within 30 minutes.

4.7 Portability

ID	Category	Description	Acceptance Criterion
NFR-PO001	Portability	The system shall be accessible across major web browsers.	The system shall function on the latest two versions of Chrome, Firefox, Safari, and Edge.
NFR-PO002	Portability	The system shall be deployable across different hosting environments.	The system shall be containerised using Docker and deployable on AWS, Azure, or Kubernetes-compatible environments.

Table 5: Non-Functional Requirements

Chapter 5: Business Rules, Constraints and Assumptions

This chapter defines the governing business policies, operational limitations, and underlying assumptions that influence the design, development, and implementation of the Mentify Digital Learning Platform.

5.1 Business Rules

ID	Business Rule
BR-001	A student may only enrol in courses that match their registered grade level unless explicitly approved by a teacher or administrator.
BR-002	Each course must have at least one assigned teacher before it can be published and made visible to students.
BR-003	Assignment submissions shall not be accepted after the deadline unless the teacher has enabled a late submission policy for that assignment.
BR-004	A student must complete all prerequisite lessons before accessing lessons configured with prerequisites by the teacher.
BR-005	Only users with the Administrator role may create, modify, or deactivate user accounts. Teachers shall not manage accounts of other teachers.
BR-006	Support team members shall have read-only access to administrative tools and shall not be permitted to modify user accounts or system configurations.
BR-007	AI-generated quiz questions must be reviewed and approved by the teacher before being published to students.
BR-008	All communication within the platform, including group chats, private messages, and discussion forums, shall comply with TIT Education's code of conduct and be subject to moderation.
BR-009	Notifications shall be delivered in real time where possible, with fallback delivery via email if the user is not actively logged into the system.

Table 6: Business Rules

5.2 Constraints

ID	Constraint
C-001	The system must be developed within the academic timeline defined by the SENG 31242 module.
C-002	The development team consists of four members (Holins, Dilakshan, Jenushan, Zaina), which limits the feature scope deliverable in the initial release.
C-003	The system shall be accessible via standard web browsers and shall not require native mobile applications in the initial release.
C-004	AI functionalities are dependent on the availability, performance, and

	cost of third-party AI/ML service providers.
C-005	The system must comply with Sri Lankan data protection regulations and institutional policies of TIT Education.
C-006	Initial deployment must align with the existing infrastructure capabilities of TIT Education.

Table 7: System Constraints

5.3 Assumptions

ID	Assumption
A-001	It is assumed that all students and teachers have access to internet-enabled devices such as laptops, tablets, or smartphones.
A-002	It is assumed that TIT Education will provide the required domain expertise, user feedback, and initial course content.
A-003	It is assumed that users possess basic digital literacy required to navigate a web-based application.
A-004	It is assumed that TIT Education will designate at least one staff member to function as the system administrator.
A-005	It is assumed that the institution will provide or procure the necessary cloud hosting infrastructure for deployment.
A-0066	It is assumed that continuous internet connectivity is required for all system operations.

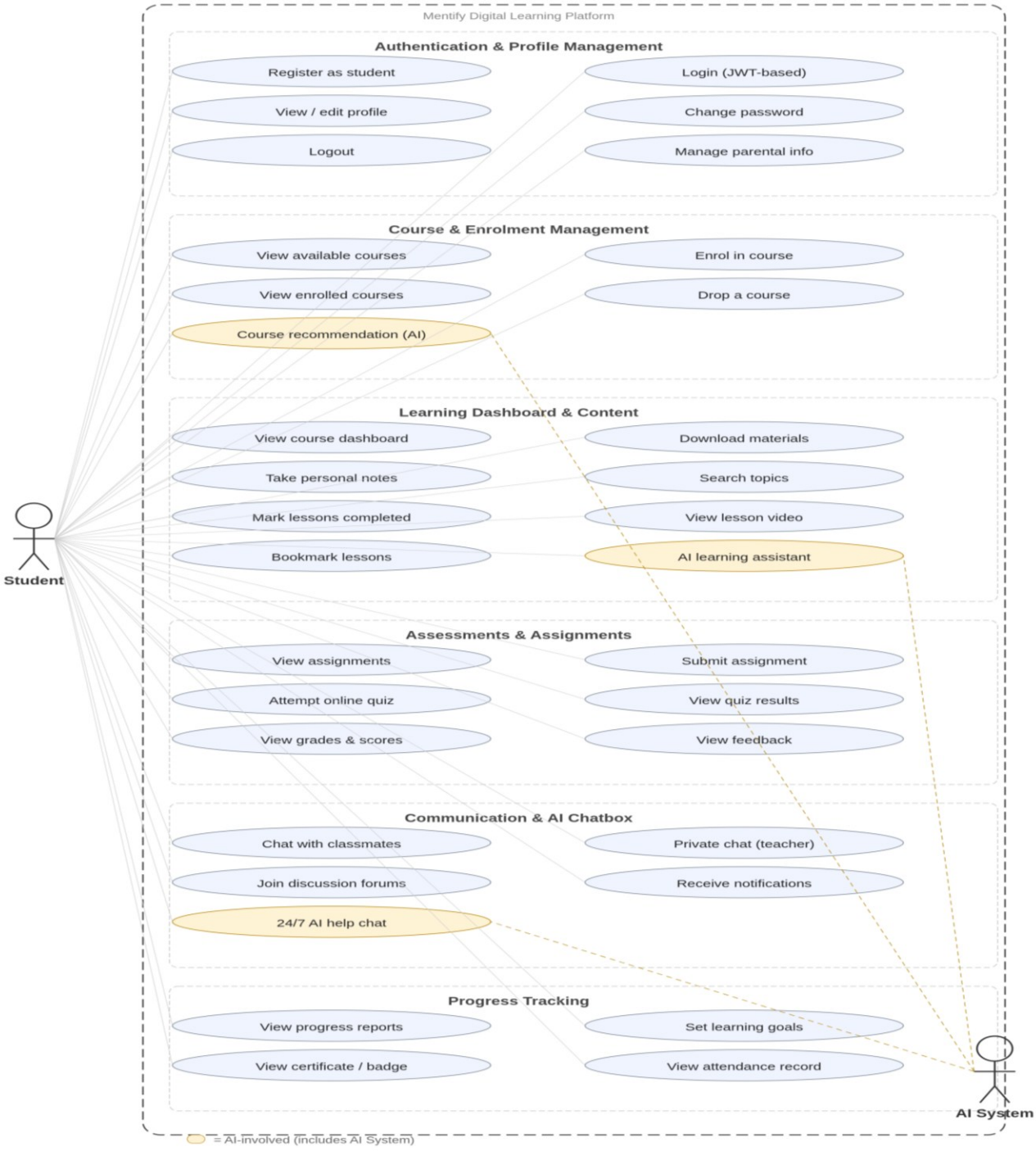
Table 8: System Assumptions

Chapter 6: Use Case Diagrams and Descriptions

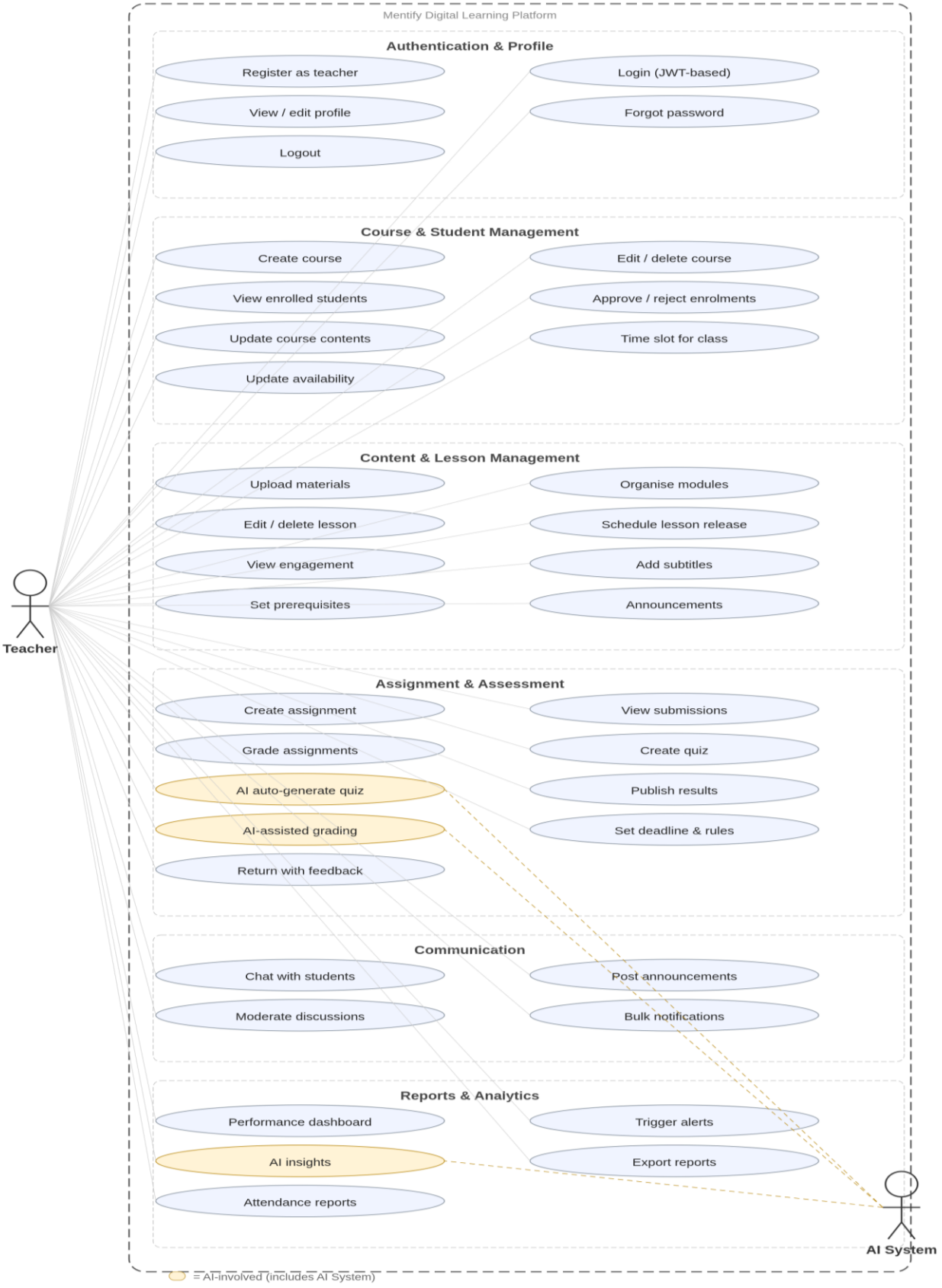
6.1 Use Case Diagrams

The use case diagrams illustrate the interactions between the primary actors and the Mentify Digital Learning Platform. The following use case diagrams are included in this chapter

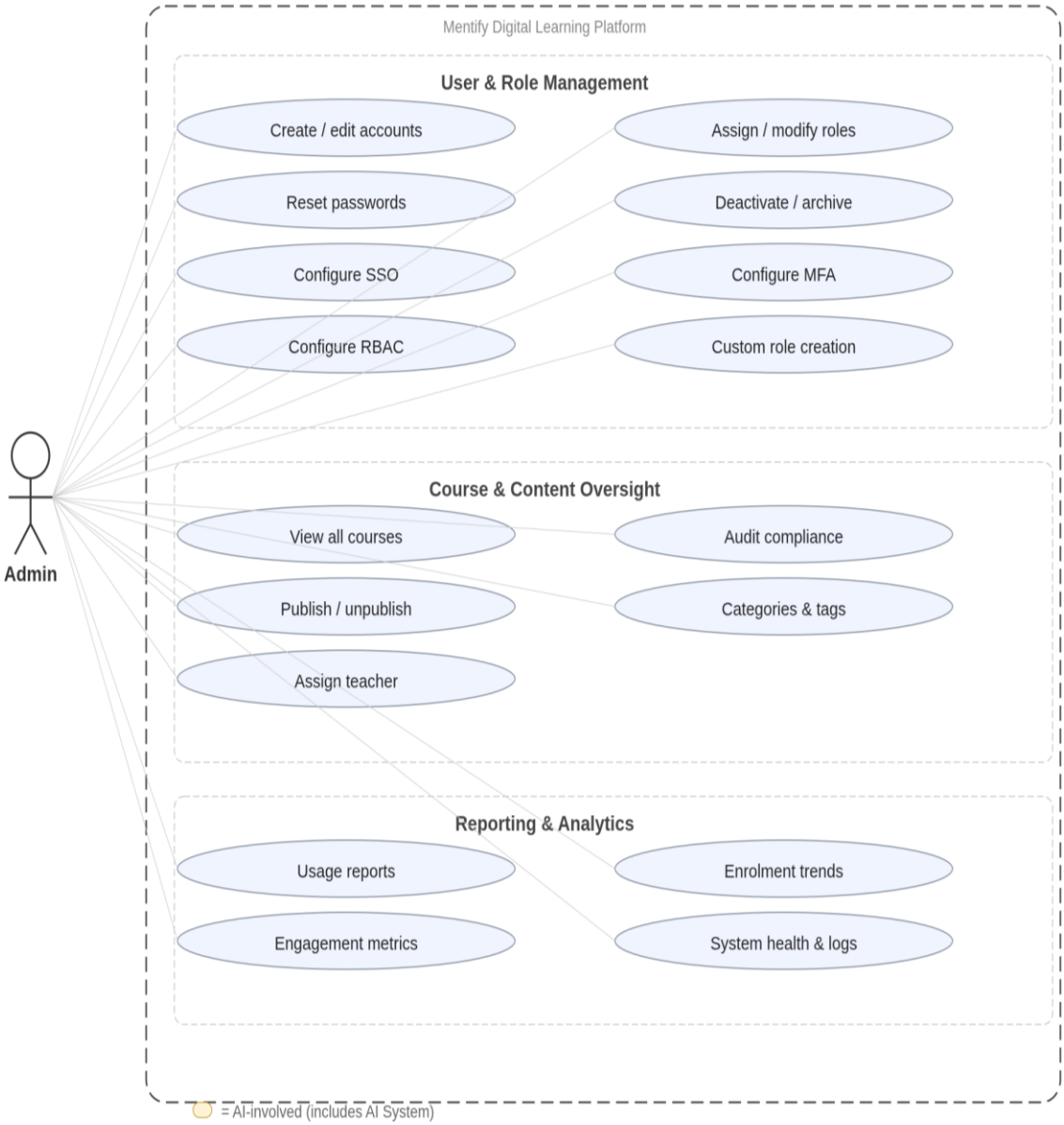
6.1.1 Student Use Case Diagram



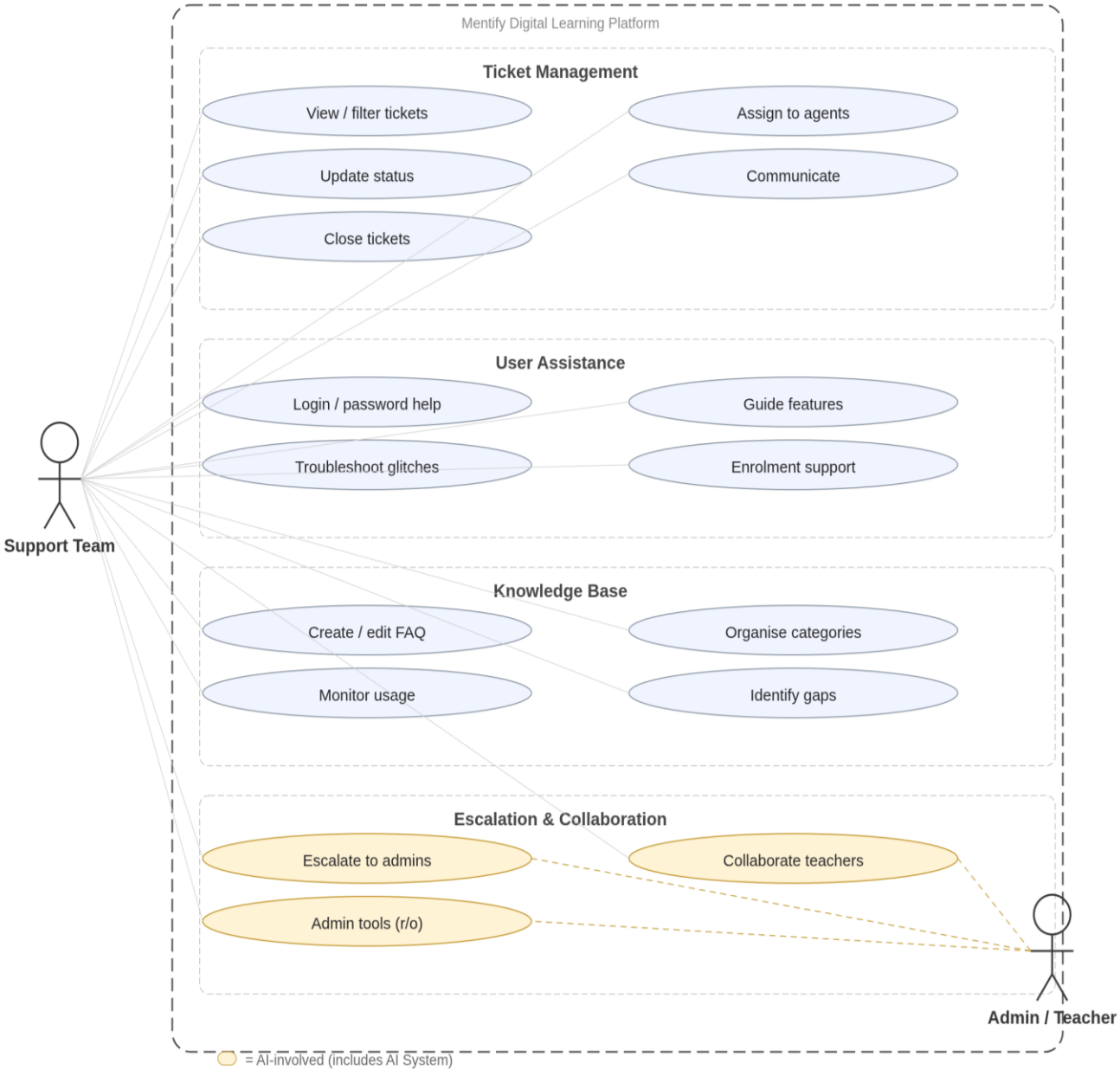
6.1.2 Teacher Use Case Diagram



6.1.3 Administrator Use Case Diagram



6.1.4 Support Team Use Case Diagram



6.2 Use Case Descriptions

The following use case descriptions present the major system interactions for the primary actors of the Mentify Digital Learning Platform.

Use Case Description: UC-S001

Field	Description
Use Case ID	UC-S001
Use Case Name	Register as Student
Primary Actor	Student
Description	A new student creates an account on the platform by providing personal details or using third-party authentication.
Preconditions	The student does not already have an account on the platform. The registration page is accessible.
Postconditions	A new student account is created and the student is redirected to the login page or dashboard.
Main Flow	1. The student navigates to the registration page. 2. The system displays the registration form. 3. The student enters their name, email, grade level, and password. 4. The student submits the form. 5. The system validates the input data. 6. The system creates the account and sends a verification email. 7. The student verifies their email. 8. The system activates the account and redirects the student to the login page.
Alternative Flow	3a. The student selects “Register with Google.” 3b. The system redirects the student to the third-party authentication provider. 3c. Upon successful authentication, the system creates the account using the received profile information.
Exception Flow	5a. If the email is already registered, the system displays the message: “An account with this email already exists.” 5b. If required fields are missing, the system highlights the missing fields and prompts the student to complete them.
Priority	High

Use Case Description: UC-S002

Field	Description
Use Case ID	UC-S002
Use Case Name	Login (JWT-based)
Primary Actor	Student
Description	The student logs into the system using email and password with JWT-based authentication.
Preconditions	The student has a registered and verified account.
Postconditions	The student is authenticated and a JWT token is issued for the session.
Main Flow	1. The student navigates to the login page. 2. The student enters their email and password. 3. The system validates the credentials. 4. The system generates a JWT access token and refresh token. 5. The system redirects the student to their dashboard.
Alternative Flow	2a. The student selects “Login with Google.” 2b. The system authenticates the student via the third-party provider and issues a JWT token upon success.
Exception Flow	3a. If the credentials are invalid, the system displays the message: “Invalid email or password.” 3b. If the account is deactivated, the system displays the message: “Your account has been deactivated. Please contact support.”
Priority	High

Use Case Description: UC-S008

Field	Description
Use Case ID	UC-S008
Use Case Name	Enrol in Course
Primary Actor	Student
Description	The student selects a course from the available courses list and enrols in it to gain access to the course content.
Preconditions	The student is logged in. The course is published and matches the student's grade level.
Postconditions	The student is enrolled in the course and it appears in their enrolled courses list.
Main Flow	1. The student navigates to the available courses page. 2. The student selects a course. 3. The system displays the course details. 4. The student clicks "Enrol." 5. The system processes the enrolment request. 6. If auto-approval is enabled, the system confirms enrolment immediately. 7. The course appears in the student's enrolled courses list.
Alternative Flow	6a. If teacher approval is required, the system sends the enrolment request to the teacher and displays the message: "Your enrolment is pending teacher approval."
Exception Flow	5a. If the student is already enrolled, the system displays the message: "You are already enrolled in this course."
Priority	High

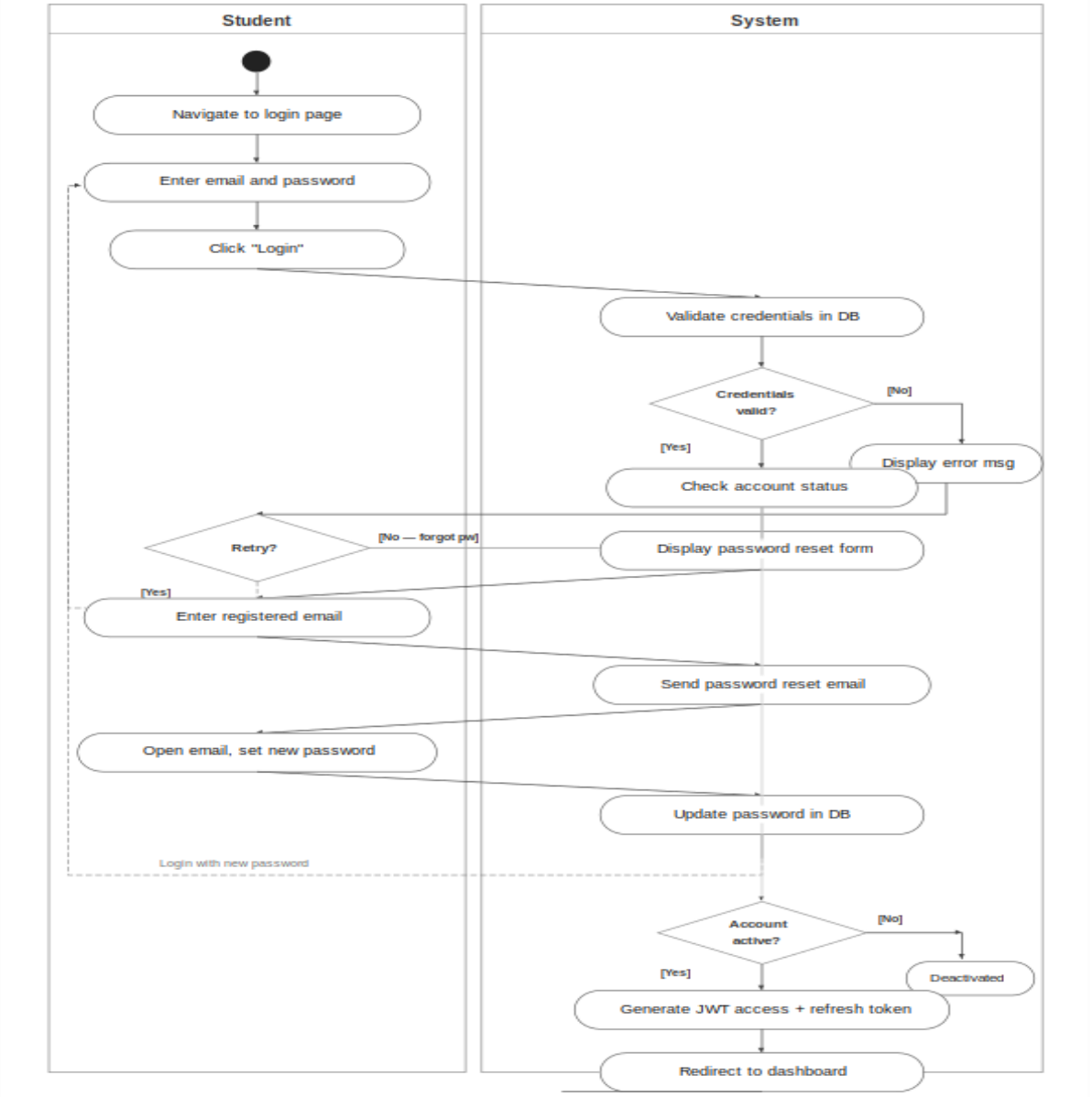
Table 9: Use Case Descriptions

Chapter 7: Activity Diagrams

Activity diagrams illustrate the dynamic workflows and operational processes of the Mentify Digital Learning Platform. These diagrams represent the sequence of actions, decision points, system validations, and outcomes involved in key user interactions and system operations.

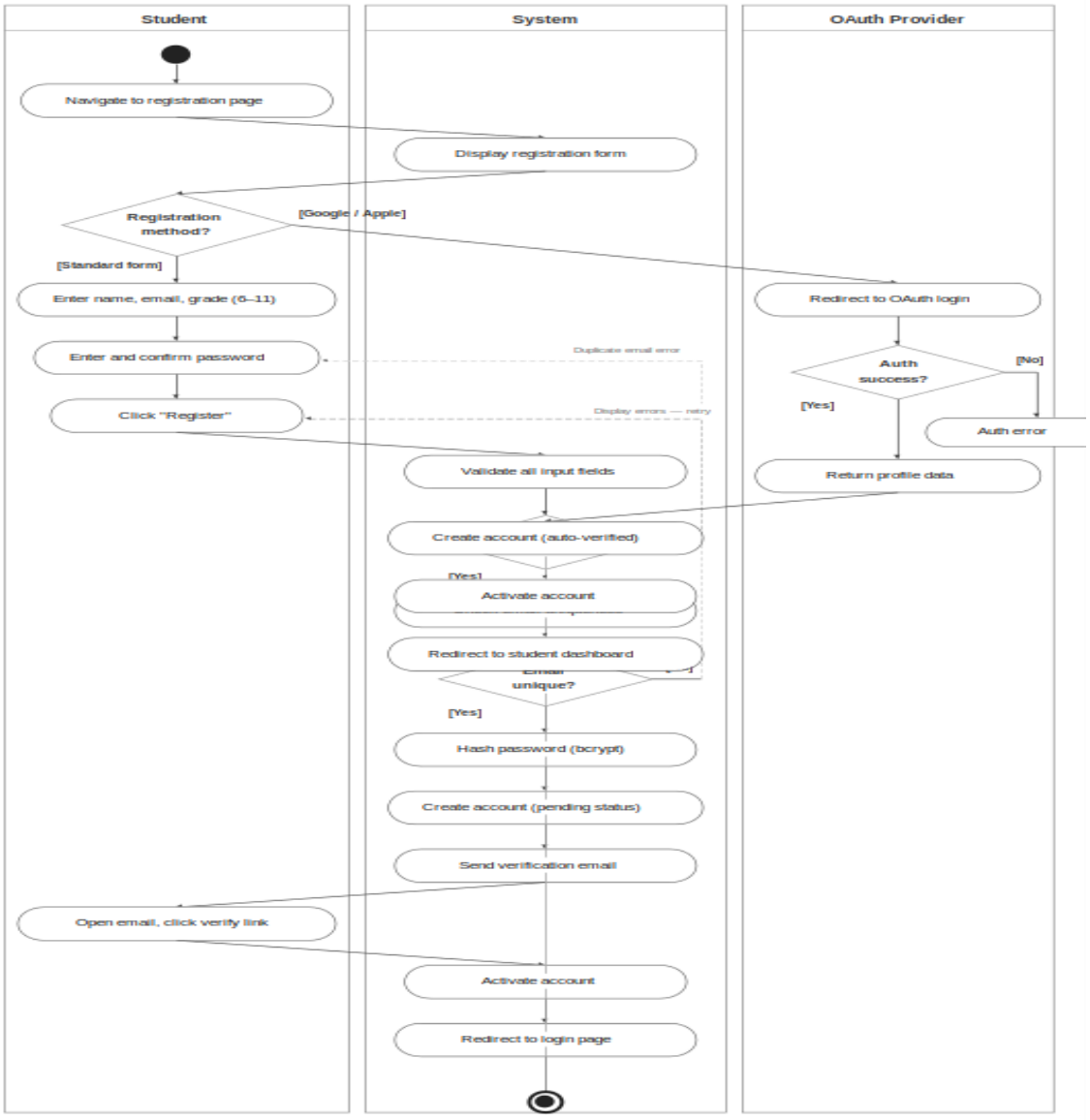
Each activity diagram provides a visual representation of system behaviour for major functional scenarios involving Students, Teachers, Administrators, Support Team members, and the AI subsystem.

7.1 Student Login (JWT-based)



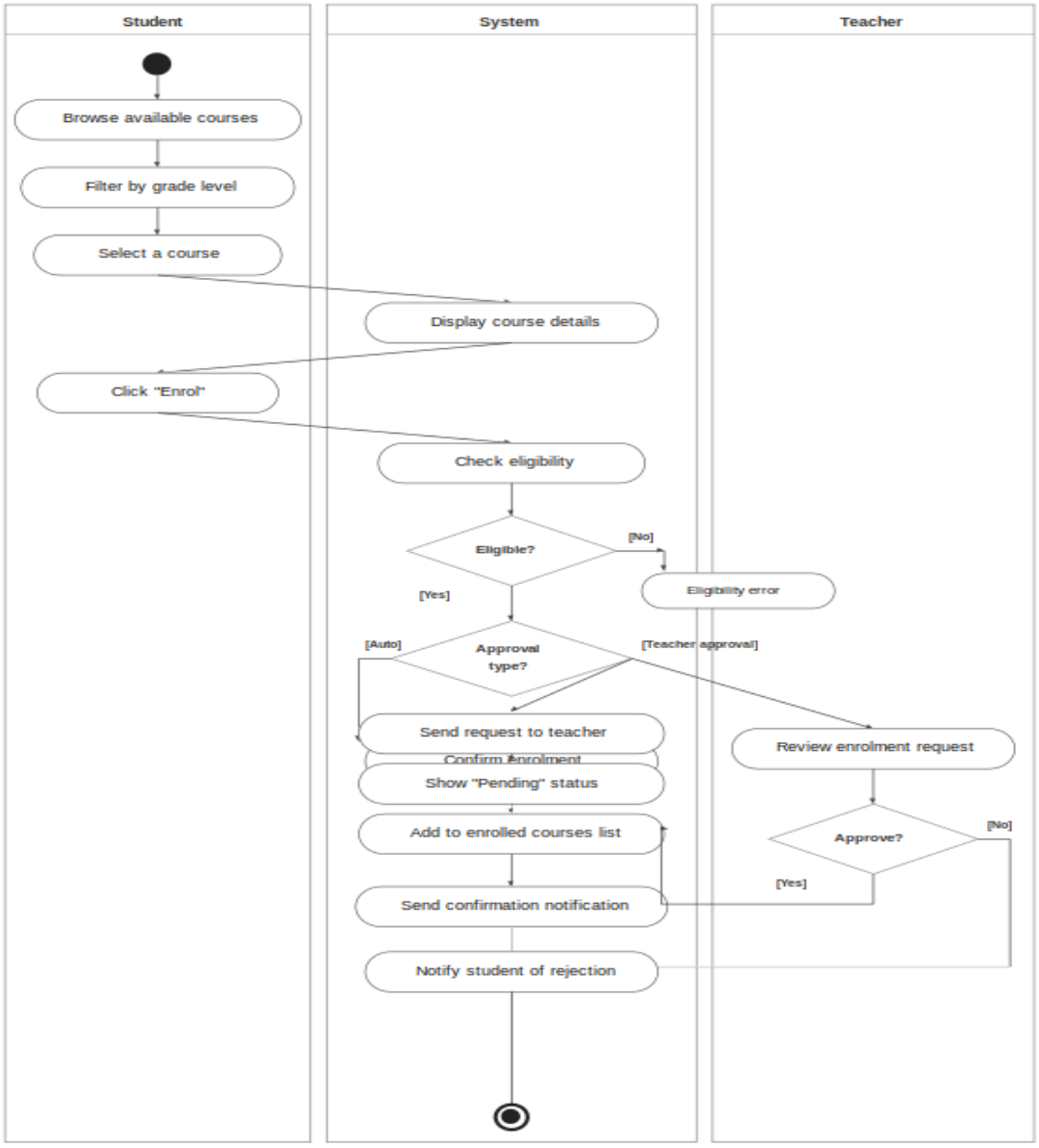
This activity diagram illustrates the student login workflow. The process begins when the student navigates to the login page and enters their email and password. The system validates the credentials. If the credentials are invalid, an error message is displayed with options to retry authentication or initiate the password recovery process. If the credentials are valid, the system verifies whether the account is active. If the account is deactivated, an appropriate message is displayed. If the account is active, the system generates a JWT access token and refresh token, establishes a session, and redirects the student to the dashboard.

7.2 Student Registration



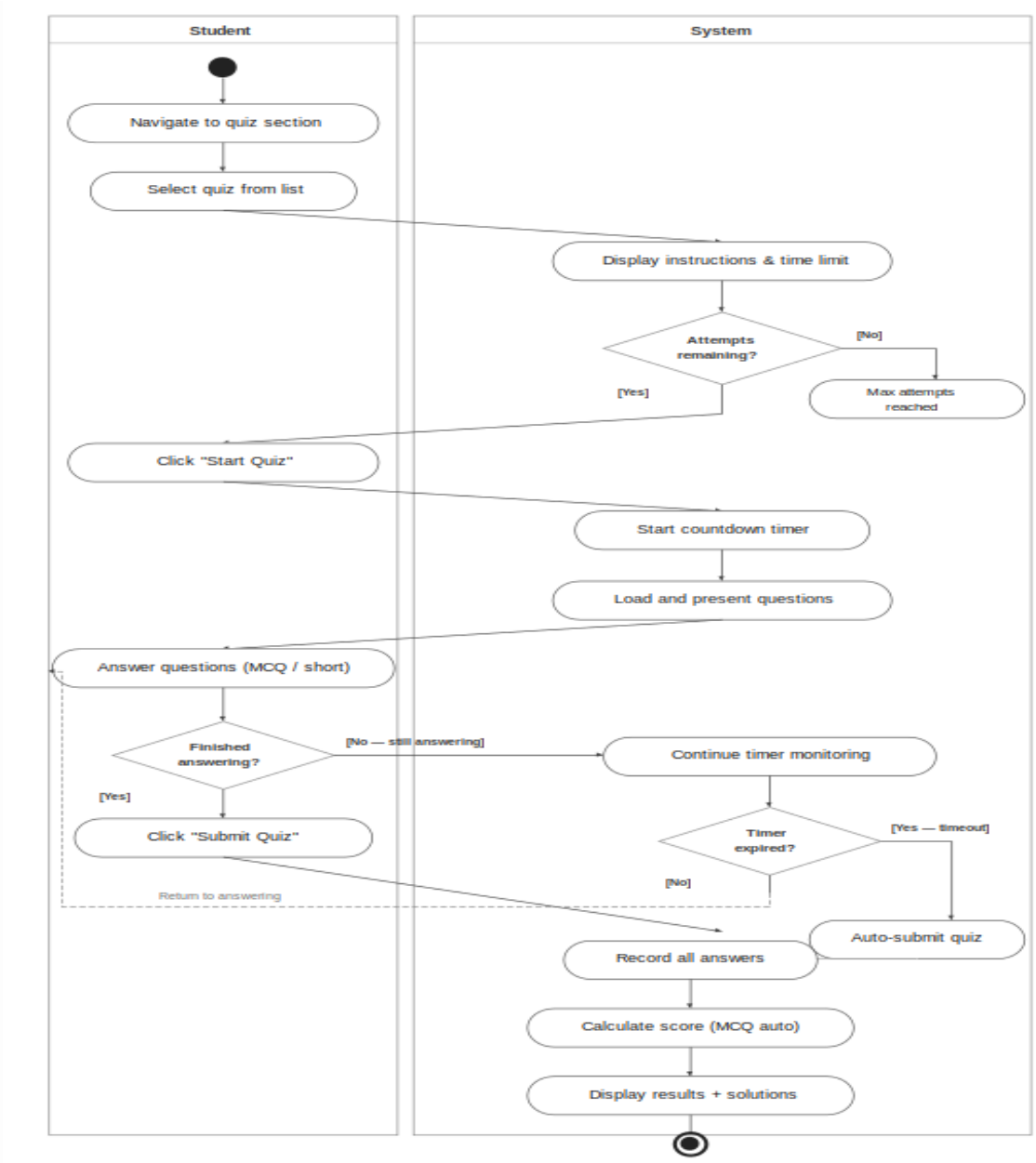
This activity diagram illustrates the student registration process. The student navigates to the registration page and selects either standard registration or third-party authentication. In the standard flow, the system validates the entered details, creates the account, and sends a verification email. The account becomes active upon successful email verification. In the third-party authentication flow, the system redirects the user to the external provider, retrieves the user profile upon successful authentication, and creates the account. Both flows converge at the account activation stage.

7.3 Enrol in Course



This activity diagram describes the course enrolment workflow. The student browses available courses, filters them based on grade level, selects a course, and initiates enrolment. The system verifies eligibility. If the student is not eligible, an error message is displayed. If eligible, the system determines whether enrolment requires teacher approval. In the case of auto-approval, enrolment is confirmed immediately. If teacher approval is required, the request is forwarded to the teacher. Upon approval, enrolment is confirmed; otherwise, the student is notified of rejection.

7.4 Attempt Online Quiz



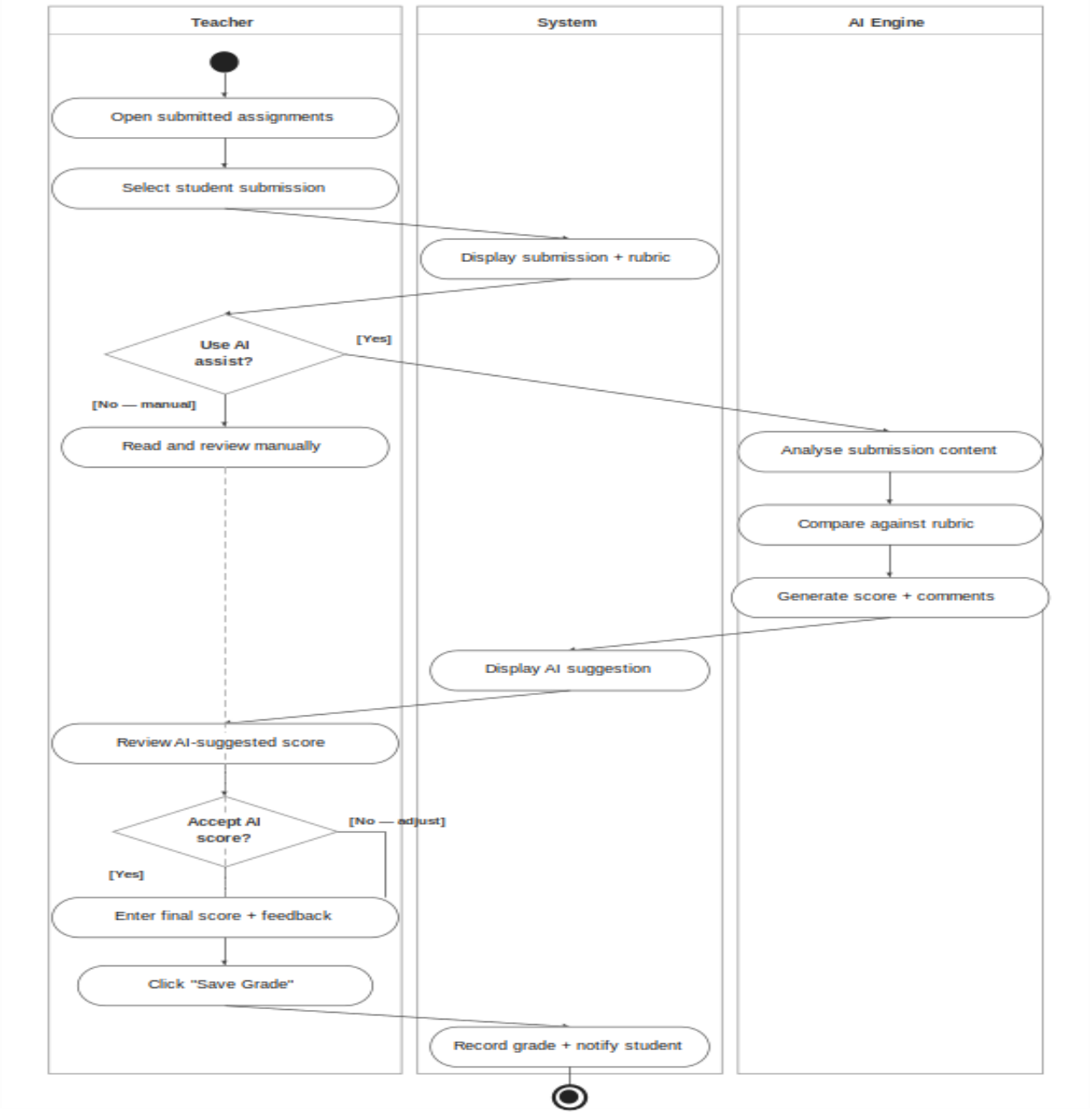
This activity diagram illustrates the quiz-taking process. The student selects a quiz and the system verifies the availability of remaining attempts. If no attempts remain, an error message is displayed. If attempts are available, the student initiates the quiz and the system starts the timer. Questions are presented sequentially or collectively, and the student submits responses. If the timer expires, the system automatically submits the quiz. Otherwise, the student submits manually. The system then evaluates objective questions and displays the results.

7.5 Teacher Creates and Publishes Quiz with AI Assistance



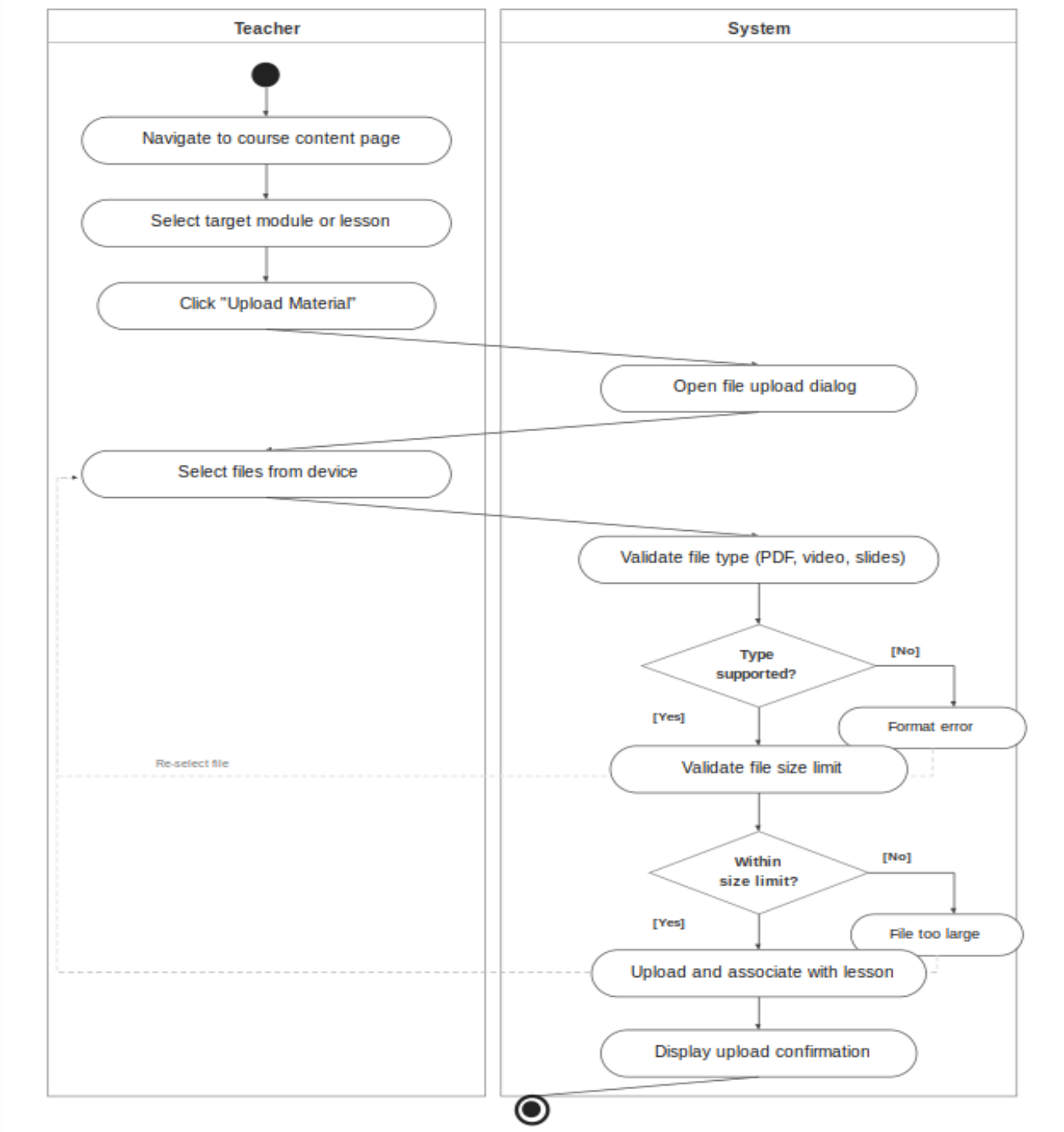
This activity diagram represents the quiz creation workflow. The teacher navigates to the assessment section, initiates quiz creation, and enters required details. The teacher may add questions manually or utilise AI-generated suggestions. In the AI flow, the system analyses course content and generates suggested questions, which the teacher reviews and modifies. Both flows converge when scoring rules are defined. The quiz is saved in draft status and subsequently published.

7.6 Teacher Grades Assignment with AI Assistance



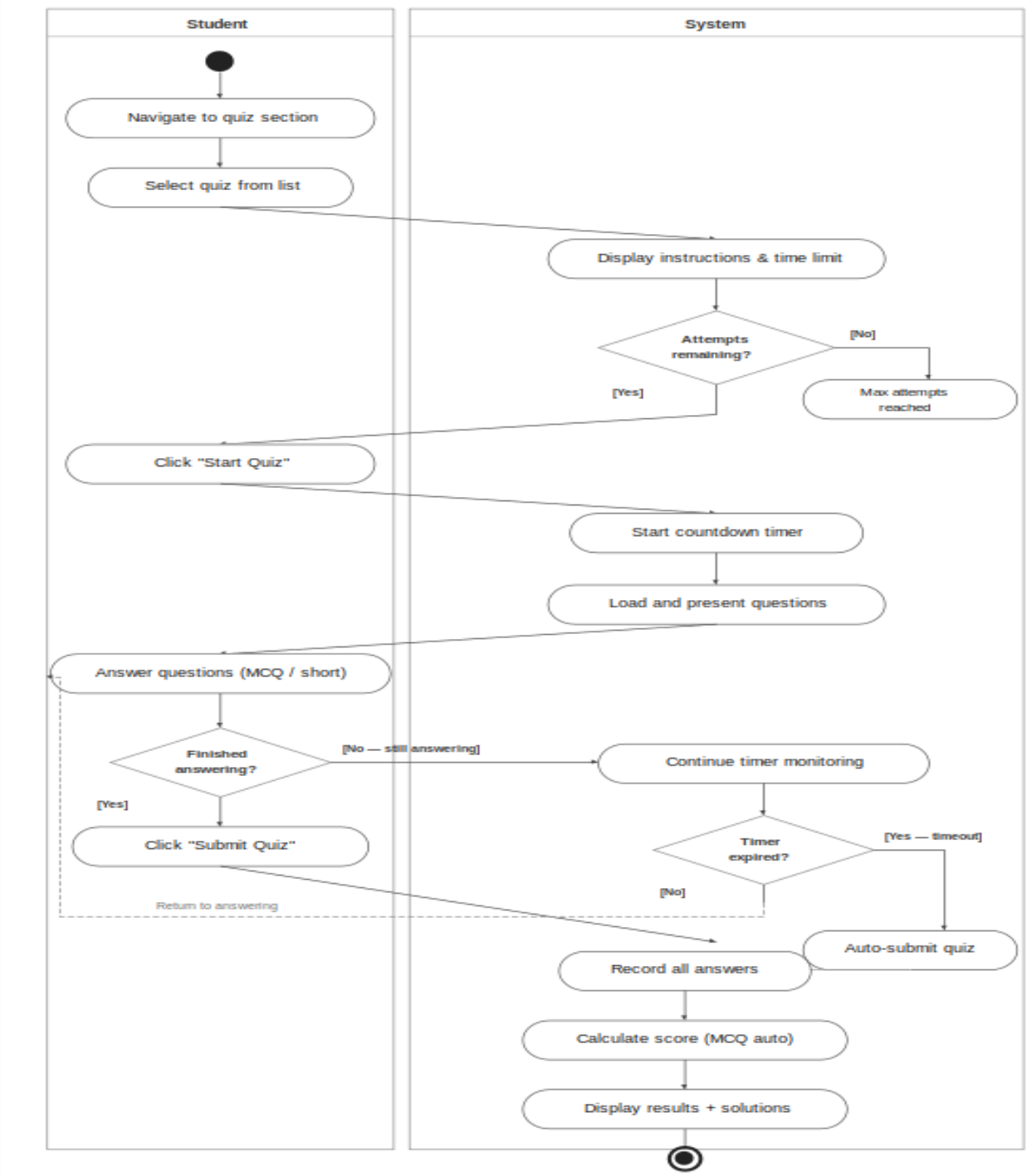
This activity diagram shows the assignment grading process. The teacher accesses submitted assignments and reviews the content. The teacher may use AI assistance to obtain suggested scores. Whether graded manually or with AI support, the process converges when the teacher finalises the score and feedback. The system records the grade and notifies the student.

7.7 Upload Learning Materials



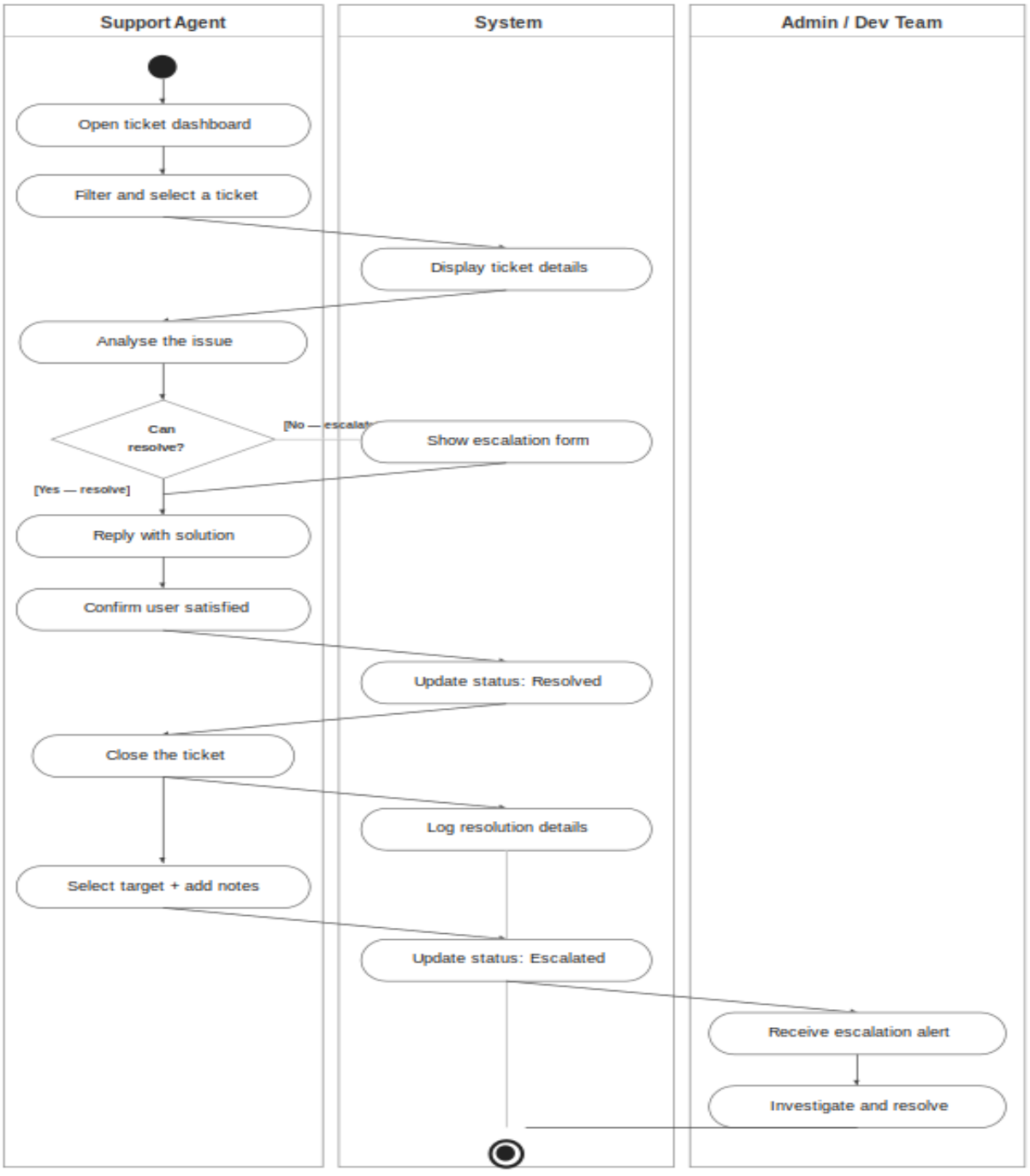
This activity diagram illustrates the content upload workflow. The teacher navigates to the course content area, selects the target module, and uploads learning materials. The system validates file type and size. If validation fails, an error is displayed. If successful, the content is uploaded and confirmation is provided.

7.8 Admin Creates User Account



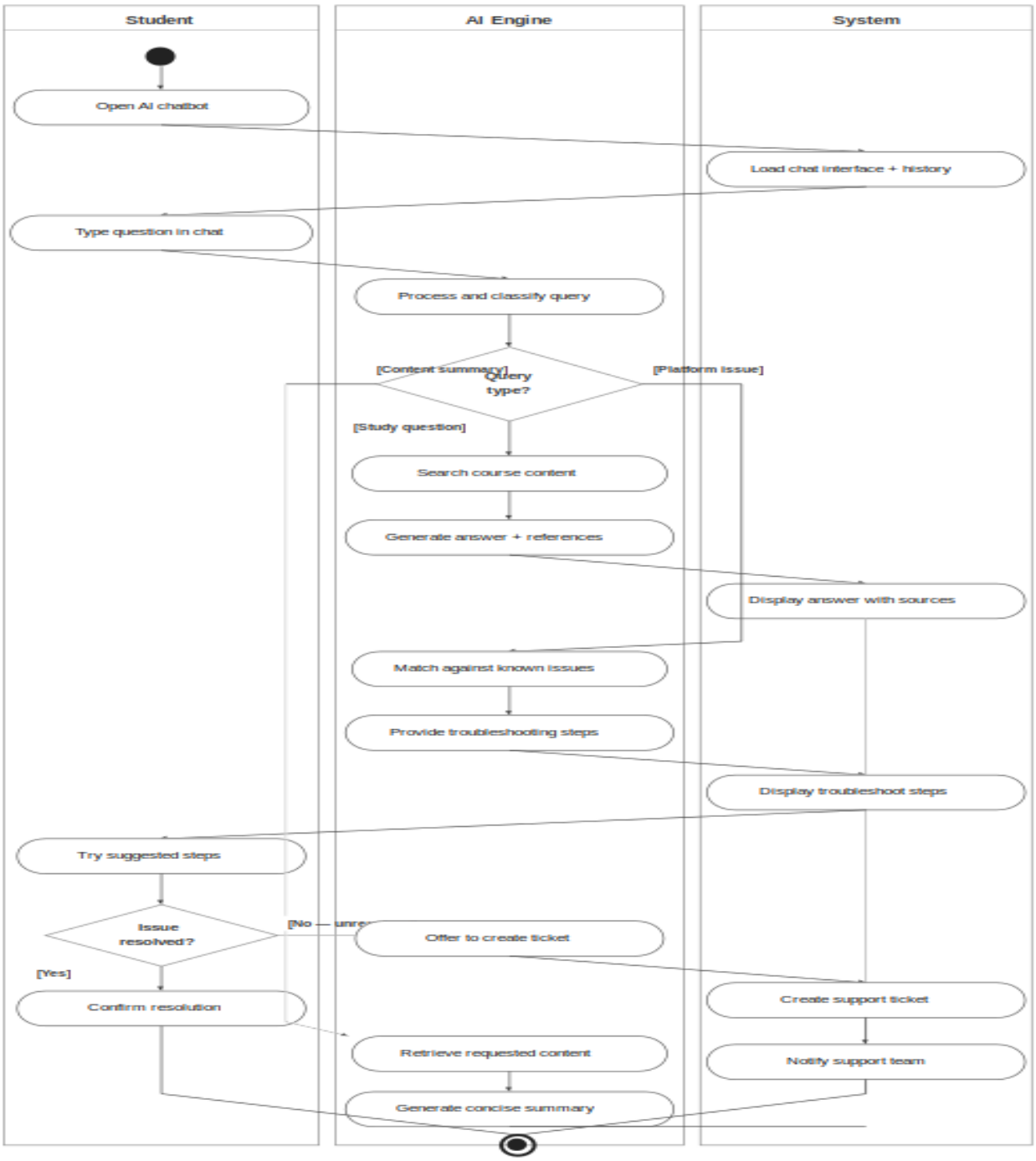
This activity diagram shows the administrative user account creation process. The administrator navigates to the user management module and enters required details. The system validates input and checks for duplicate records. If validation fails, errors are displayed. If validation succeeds, the account is created and a welcome email is sent.

7.9 Support Team Handles Ticket



This activity diagram describes the support ticket lifecycle. The support agent reviews the issue and determines whether it can be resolved. If resolved, the ticket is updated and closed. If unresolved, the agent escalates the issue to the administrator or development team. The system updates the ticket status and sends notifications.

7.10 AI Help Chatbot



This activity diagram represents the AI chatbot interaction process. The student submits a query through the chatbot interface. The AI system analyses the query and categorises it as a study question, platform issue, or content request. Based on classification, the system generates an appropriate response. If the issue remains unresolved, the chatbot offers to create a support ticket.

Chapter 8: External Interface Requirements

This chapter describes the external interfaces of the Mentify Digital Learning Platform, including user interface characteristics, hardware interactions, software integrations, and communication protocols required for system operation.

8.1 User Interface Requirements

The system shall provide a modern, responsive, web-based user interface that is intuitive, accessible, and aligned with contemporary usability standards. The interface shall adopt a clean and minimalist design with consistent colour theming that reflects the branding guidelines of TIT Education.

The system interface shall be fully responsive and capable of adapting seamlessly across desktop, tablet, and mobile devices. Accessibility requirements shall comply with WCAG 2.1 Level AA standards, including appropriate colour contrast ratios, keyboard navigation support, screen reader compatibility, and alternative text for all non-text content.

The dashboard shall utilise visual elements such as progress indicators, charts, and card-based layouts to present information in a clear and structured manner. All input forms shall implement client-side validation with descriptive and user-friendly error messages. Navigation shall be role-based, ensuring that users only view features and menu options relevant to their assigned roles.

8.2 Hardware Interface Requirements

The Mentify Digital Learning Platform is designed as a cloud-hosted web application and therefore does not require direct integration with specialised hardware components. However, the system shall support interaction through standard input devices, including keyboards, mice, and touch-enabled interfaces.

The platform shall be compatible with display devices ranging from small mobile screens (approximately 320 pixels in width) to high-resolution displays up to 4K resolution. The system shall support file uploads from locally connected storage devices available on the user's machine.

8.3 Software Interface Requirements

The system shall integrate with external software services and application programming interfaces to support authentication, AI-powered functionality, storage, messaging, and data management.

Third-party authentication shall be supported through integration with Google OAuth 2.0. AI-driven functionalities, including content summarisation, automated quiz generation, grading assistance, course recommendations, and chatbot interaction, shall be implemented through integration with AI service providers such as OpenAI or equivalent platforms.

The system shall interface with email delivery services via SMTP-based providers such as SendGrid or AWS Simple Email Service to facilitate transactional communications, including account verification, password reset notifications, and system alerts.

Cloud storage services such as Amazon S3 or Google Cloud Storage shall be utilised for storing multimedia learning materials, assignment submissions, and related content. The platform shall interface with a relational database management system, such as PostgreSQL, for structured data storage, and may utilise a NoSQL database such as MongoDB for storing chat logs and user activity data.

Additionally, the system shall integrate with a push notification service to enable real-time in-app notifications and mobile push messaging.

8.4 Communication Interface Requirements

All communication between client and server components shall be conducted over secure HTTPS connections using TLS version 1.2 or higher. The system shall expose RESTful application programming interfaces that use JSON as the primary data exchange format.

Real-time communication features, including messaging and notifications, shall be implemented using the WebSocket protocol to ensure low-latency data transmission. The system APIs shall conform to REST architectural principles and utilise appropriate HTTP status codes for all responses.

File transfer operations shall support multipart/form-data encoding for efficient handling of uploaded content. The system shall also implement rate-limiting mechanisms across API endpoints to prevent misuse, ensure service stability, and enhance overall security.

Chapter 9: Data Requirements

This chapter describes the types of data that shall be managed, processed, stored, and protected by the Mentify Digital Learning Platform. The system shall handle multiple categories of data essential for learning delivery, assessment, communication, analytics, and support operations.

9.1 User Data

The system shall store and manage user-related data, including personal identification information such as name, email address, grade level, and assigned role. Authentication-related data shall include securely hashed passwords and issued JWT tokens. Additional profile information may include profile photographs, contact details, and parental contact information where applicable.

9.2 Course Data

The system shall maintain course-related data, including course metadata such as course name, description, grade level, category, and associated tags. It shall store structured content organisation, including modules and lessons, along with associated learning materials such as documents, multimedia files, and videos. The system shall also maintain prerequisite configurations and scheduled release timelines for lessons and modules.

9.3 Assessment Data

The platform shall manage assessment-related data, including assignment specifications such as instructions, submission deadlines, and submission rules. It shall store quiz configurations, including question sets, answer options, scoring logic, and time constraints. Student submission data shall include uploaded files, textual responses, and submission timestamps. The system shall also record grades, performance scores, and feedback provided by teachers or generated through AI assistance.

9.4 Communication Data

The system shall store communication-related data, including group and private chat messages, discussion forum content, announcements, and system-generated notifications. This data shall support collaborative learning and information dissemination across the platform.

9.5 Analytics Data

The system shall collect and maintain analytics-related data, including student engagement metrics such as lesson views, time spent on learning activities, completion rates, and attendance records. Performance analytics shall include score trends, learning progress indicators, and system usage logs used for monitoring platform effectiveness.

9.6 Support Data

Support-related data shall include support ticket information such as ticket status, priority level, issue category, and resolution notes. The system shall also maintain knowledge base content and escalation history records to support efficient issue management.

9.7 Data Security and Retention

All personal and sensitive data shall be encrypted at rest using AES-256 encryption standards. Data retention policies shall comply with the institutional policies of TIT Education as well as applicable data protection regulations. Automated database backups shall be performed at intervals of 24 hours and retained for a minimum period of 30 days to ensure data recoverability and system reliability.

Chapter 10: Alternative Solutions Considered

During the system planning and requirement analysis phase, several alternative implementation approaches were evaluated to determine the most suitable solution for developing the Mentify Digital Learning Platform.

10.1 Adoption of an Existing Learning Management System

One potential approach considered was the adoption of an existing Learning Management System such as Moodle, Canvas, or Google Classroom. These platforms provide pre-built functionalities for course management, assessment delivery, and communication features. However, they lack the advanced AI-driven personalisation capabilities required by TIT Education. Furthermore, these platforms offer limited flexibility for customisation to align with the institution's specific workflows and pedagogical strategies. Proprietary platforms also involve recurring licensing costs. Although Moodle is open-source, it would require extensive custom development to support the desired AI-based functionality and tailored system features.

10.2 Low-Code / No-Code Development Platforms

Another alternative involved the use of low-code or no-code development platforms such as Bubble or AppSheet. This approach could potentially accelerate the development timeline and reduce initial implementation complexity. However, such platforms impose constraints on system performance, scalability, and the integration of advanced AI capabilities. Additionally, reliance on proprietary development environments may introduce long-term challenges in maintainability, system portability, and vendor dependency.

10.3 Custom System Development (Selected Approach)

The selected approach for the Mentify Digital Learning Platform is full custom system development. This approach enables complete control over system functionality, user experience design, AI integration, and architectural scalability. Although custom development requires greater initial development effort, it provides the flexibility needed to address TIT Education's requirement for a student-centred, AI-powered digital learning environment. Moreover, a custom-built solution allows future expansion and scalability beyond the institutional context.

Chapter 11: Feasibility Study

This chapter evaluates the feasibility of developing and implementing the Mentify Digital Learning Platform across technical, economic, operational, and schedule dimensions.

11.1 Technical Feasibility

The proposed system is considered technically feasible. The selected technology stack, comprising React.js for the frontend, Spring Boot for the backend, PostgreSQL for database management, and cloud hosting through AWS or an equivalent provider, is mature, well-documented, and widely supported within the software development ecosystem.

AI-driven functionalities can be effectively implemented through integration with established AI service providers such as OpenAI. Core architectural components including JWT-based authentication, Role-Based Access Control (RBAC), and WebSocket-based real-time communication are supported by robust frameworks and libraries.

Furthermore, the development team possesses the required competencies in web application development, database management, and API integration, ensuring that the technical requirements of the system can be successfully addressed.

11.2 Economic Feasibility

The project demonstrates strong economic feasibility as it is being developed within an academic context, thereby significantly reducing labour-related costs. Infrastructure expenses can be managed effectively through the utilisation of cloud provider free tiers and educational support programmes such as AWS Educate and Google Cloud for Education.

Primary anticipated costs include cloud hosting services, AI API usage, and domain registration. These expenses are expected to remain within a manageable budget. From the perspective of TIT Education, the implementation of this system eliminates the need for costly commercial Learning Management System subscriptions and reduces administrative workload through automation, resulting in long-term cost savings.

11.3 Operational Feasibility

Operational feasibility is supported by the positive reception of the system concept among the staff of TIT Education, who have indicated readiness to adopt the proposed platform. The system is designed with a strong emphasis on usability, requiring only basic digital literacy from its users.

The availability of training materials and an integrated knowledge base will facilitate user onboarding and system adoption. Additionally, the inclusion of an AI-powered chatbot capable of providing continuous assistance reduces the operational workload on the support team. The system is intended to operate alongside existing manual processes during the transition phase, enabling gradual adoption without significant operational disruption.

11.4 Schedule Feasibility

The development of the Mentify Digital Learning Platform aligns with the academic schedule defined by the SENG 31242 module. The project is structured into phased development stages, prioritising essential system components such as authentication, course management, content delivery, and assessment functionalities for the initial release.

Advanced features, including AI-driven personalisation, analytics dashboards, and comprehensive reporting capabilities, are planned for subsequent iterations. The development team, consisting of four members, has distributed responsibilities to support parallel development activities. The adoption of Agile development practices, including two-week sprint cycles, facilitates continuous progress monitoring and supports timely project completion.

Chapter 12: Risk Analysis

This chapter identifies potential risks associated with the development, deployment, and operation of the Mentify Digital Learning Platform. Each risk is evaluated based on its likelihood and impact, and appropriate mitigation strategies are defined to minimise potential adverse effects.

Risk ID	Description	Like lihood	Impact	Risk Level	Mitigation Strategy
R-001	AI API service outages or rate limiting may disrupt AI features such as summarisation, quiz generation, and chatbot functionality.	Medium	High	High	Implement graceful degradation so that the platform remains functional without AI features. Cache AI responses where feasible and evaluate multiple AI providers for fallback support.
R-002	Scope creep due to evolving client requirements may delay the project timeline.	High	High	High	Establish a formal change request process. Prioritise features using the MoSCoW methodology and maintain a fixed scope for each sprint.
R-003	Team member unavailability due to academic or personal commitments.	Medium	Medium	Medium	Cross-train team members across modules. Maintain comprehensive documentation and utilise version control with frequent commits.
R-004	Data breach or security vulnerability may expose sensitive student information.	Low	High	High	Conduct regular security audits, implement encryption for data at rest and in transit, follow

					OWASP security guidelines, and enforce multi-factor authentication for administrative accounts.
R-005	Poor user adoption by teachers and students unfamiliar with digital learning platforms.	Medium	Medium	Medium	Design an intuitive user interface, provide onboarding tutorials, maintain a comprehensive knowledge base, and conduct structured user training sessions.
R-006	Performance degradation under high concurrent user load during peak usage periods.	Medium	High	High	Implement load balancing and horizontal scaling. Conduct stress testing prior to deployment. Utilise content delivery networks for static resources and optimise database queries using indexing.
R-007	Failure of third-party services such as authentication providers, email services, or cloud storage.	Low	Medium	Medium	Implement fallback authentication mechanisms, use redundant email service providers, and design retry mechanisms for failed operations.
R-008	Insufficient testing coverage leading to undetected defects in production.	Medium	Medium	Medium	Maintain code coverage above 80%, implement automated unit, integration, and end-to-end testing, and conduct user acceptance testing with TIT Education prior to deployment.

R-009	Database corruption or data loss resulting from system failure.	Low	High	Medium	Implement automated daily backups with geographic redundancy, enable point-in-time recovery, and regularly test restoration procedures.
R-010	Regulatory compliance issues related to student data handling.	Low	High	Medium	Align with institutional data policies, implement data anonymisation for analytics, obtain parental consent where required, and document all data processing activities.

Table 13: Risk Register

Chapter 13: References

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